

Mohsin Furkh Dar

Assistant Professor (Research Faculty track) • School of Computer Science, UPES, Dehradun, India

Medical Image Analysis • Ultrasound • Uncertainty-Aware and Interpretable Deep Learning

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RESEARCH PROFILE

I develop deep learning methods for ultrasound image analysis that treat ambiguity – at lesion boundaries and across datasets – as a signal rather than noise, so that models stay reliable on the small, heterogeneous datasets typical of point-of-care imaging. My doctoral work introduced a fuzzy rough set loss (*Computerized Medical Imaging and Graphics*, 2026) and an adaptive ensemble loss (*Medical & Biological Engineering & Computing*, 2025) for segmentation, and efficient, interpretable classifiers for breast ultrasound (*Neural Processing Letters*, 2023; *Image and Vision Computing*, 2024; *Biomedical Signal Processing and Control*, 2026).

At UPES I extended this work to breast, thyroid and fetal ultrasound as first and corresponding author of MSCT-Trans (*Ultrasound in Medicine & Biology*, 2026), written without my doctoral advisor. Next, I plan to extend uncertainty-aware learning to maternal–fetal ultrasound.

Record: 9 publications since 2022 (7 as first author; 4 in SJR Q1 journals); 115 citations and h-index 4 (Google Scholar, September 2026); code released publicly for 6 papers; reviewer for 8 journals, including IEEE TMI and IEEE JBHI.

9	7	115	4	4	8
PUBLICATIONS	FIRST AUTHOR	CITATIONS	H-INDEX	SJR Q1 PAPERS	JOURNALS REVIEWED FOR

EDUCATION

Nov 2020 – Jul 2026	Ph.D. in Computer Science DEGREE AWARDED 28 JULY 2026 University of Hyderabad, India <i>Thesis: Advances in Deep Learning for Medical Image Segmentation and Classification</i> - Advisor: Dr. Avatharam Ganivada, School of Computer and Information Sciences - Funded by the UGC Junior Research Fellowship (All India Rank 53)
Sep 2017 – Mar 2019	M.Phil. in Computer Science Mewar University, India <i>Thesis: Performance Comparison of Face Detection and Recognition Algorithms</i>
Mar 2013 – Jun 2016	Master of Computer Applications (MCA) University of Kashmir, India <i>Project: SMS Intimation System for Online Leave Management</i>
Mar 2010 – Jan 2013	B.Sc. (Mathematics, Physics, Information Technology) University of Kashmir, India

ACADEMIC APPOINTMENTS

Jul 2025 – Present	Assistant Professor School of Computer Science, UPES, Dehradun, India - Moved to the Research Faculty track in January 2026 - First and corresponding author of MSCT-Trans (<i>Ultrasound in Medicine & Biology</i>, 2026), written at UPES without my doctoral advisor - Teach B.Tech. and M.Tech. Computer Science courses; supervise B.Tech. and M.Tech. projects in machine learning and computer vision
Jan 2022 – Dec 2024	Teaching Assistant School of Computer and Information Sciences, University of Hyderabad, India - Delivered tutorial and laboratory sessions for graduate courses in deep learning and computer vision - Designed and evaluated programming assignments; mentored IMTech and M.Tech. students on course projects

Jan 2021 – Dec 2022

System Administrator, Artificial Intelligence Lab

School of Computer and Information Sciences, University of Hyderabad, India

- Managed GPU workstations and server infrastructure supporting the AI research group
- Configured and optimised deep learning software environments for lab-wide research use

May 2020 – Nov 2020

Assistant Professor

Government Degree College Uri, Baramulla, J&K, India

- Taught undergraduate Computer Science courses; developed material for programming and data structures

GRANTS AND FUNDING

Institution of Eminence (IoE) International Travel Grant – University of Hyderabad 2024

INR 1,00,000 covering registration, international travel and per diem (Sanction Order No. UoH-IoE/Travel/24/375, 24 July 2024), to present *Dynamic Weight-Adjusted Ensemble Loss for Enhanced Medical Image Segmentation* at ICCNet-2024, Manchester, United Kingdom, 17–18 October 2024.

RESEARCH PUBLICATIONS

Author name in bold. Quartiles per Scimago (SJR); citation counts per Google Scholar, 23 September 2026.

JOURNAL ARTICLES – MEDICAL IMAGE ANALYSIS

- [1] **M. F. Dar** and S. Mukhtar, “MSCT-Trans: A multi-scale convolutional neural network token transformer for interpretable ultrasound image classification,” *Ultrasound in Medicine & Biology*, vol. 52, no. 11, pp. 2623–2638, 2026. doi: 10.1016/j.ultrasmedbio.2026.07.032 **SCIE · Q1** **First & corresponding author**
- [2] **M. F. Dar** and A. Ganivada, “Saliency-guided AttentionNet: Dual-branch deep learning for breast ultrasound classification,” *Biomedical Signal Processing and Control*, vol. 127, art. 111194, 2026. doi: 10.1016/j.bspc.2026.111194 **SCIE · Q1**
- [3] **M. F. Dar** and A. Ganivada, “Fuzzy rough set loss for deep learning-based precise medical image segmentation,” *Computerized Medical Imaging and Graphics*, vol. 128, art. 102716, 2026. doi: 10.1016/j.compmedimag.2026.102716 **SCIE · Q1**
3 citations
- [4] **M. F. Dar** and A. Ganivada, “Adaptive ensemble loss and multi-scale attention in breast ultrasound segmentation with UMA-Net,” *Medical & Biological Engineering & Computing*, vol. 63, no. 6, pp. 1697–1713, 2025. doi: 10.1007/s11517-025-03301-5 **SCIE · Q2** **14 citations**
- [5] **M. F. Dar** and A. Ganivada, “Deep learning and genetic algorithm-based ensemble model for feature selection and classification of breast ultrasound images,” *Image and Vision Computing*, vol. 146, art. 105018, 2024. doi: 10.1016/j.imavis.2024.105018 **SCIE · Q1** **38 citations**
- [6] **M. F. Dar** and A. Ganivada, “EfficientU-Net: A novel deep learning method for breast tumor segmentation and classification in ultrasound images,” *Neural Processing Letters*, vol. 55, no. 8, pp. 10439–10462, 2023. doi: 10.1007/s11063-023-11333-x **SCIE · Q2** **48 citations**

CONFERENCE PROCEEDINGS

- [7] **M. F. Dar** and A. Ganivada, “Dynamic weight-adjusted ensemble loss for enhanced medical image segmentation,” in *Proc. 4th International Conference on Computing and Communication Networks (ICCCNet-2024)*, Manchester, UK, Lecture Notes in Networks and Systems. Singapore: Springer Nature, 2025, pp. 1–12. doi: 10.1007/978-981-96-3250-3_1 **1 citation**

OTHER CONTRIBUTIONS (OUTSIDE MEDICAL IMAGING)

- [8] A. N. Alhaj, N. D. Patel, A. Singh, R. K. Bondugula, **M. F. Dar**, and J. Ahamed, “Design and analysis of a robust security layer for software defined network framework,” *International Journal of Sensor Networks*, vol. 46, no. 1, pp. 1–14, 2024. doi: 10.1504/IJSNET.2024.141613 **8 citations**
- [9] S. Mukhtar, **M. F. Dar**, and A. Kaur, “Latent fingerprint enhancement and matching using intuitionistic type-2 fuzzy,” *International Journal of Artificial Intelligence and Soft Computing*, vol. 7, no. 4, pp. 313–328, 2022. doi: 10.1504/IJAISC.2022.130558 **3 citations**

EARLY WORK (M.PHIL.)

- [10] **M. F. Dar** and S. Dixit, “Performance comparison of face detection and recognition algorithms,” *International Journal of Science and Research*, vol. 8, no. 1, pp. 986–994, 2019. doi: 10.21275/ART20194439

RESEARCH INTERESTS AND SCHOLARLY SERVICE

RESEARCH INTERESTS

- Ambiguity- and uncertainty-aware deep learning for ultrasound image analysis
- Loss function design: fuzzy rough set and adaptive ensemble objectives
- Interpretable models: saliency-guided attention and multi-scale token transformers
- Medical image segmentation and classification; cross-dataset generalisation

- Soft computing: fuzzy-rough learning and evolutionary feature selection

PEER REVIEW – JOURNALS

- IEEE Transactions on Medical Imaging
- IEEE Journal of Biomedical and Health Informatics
- Information Fusion (Elsevier)
- Image and Vision Computing (Elsevier)
- Engineering Applications of Artificial Intelligence (Elsevier)
- Neural Computing and Applications (Springer)
- Multimedia Tools and Applications (Springer)
- Journal of Clinical Ultrasound (Wiley)

OPEN SCIENCE

- Code for six publications released publicly at github.com/MohsinFurkh (EfficientU-Net, GA ensemble, UMA-Net, fuzzy rough set loss, Saliency-Guided AttentionNet, MSCT-Trans)

HONOURS

2019 **UGC NET with Junior Research Fellowship (JRF)**, Computer Science & Applications – **All India Rank 53**

PRESENTATIONS AND PROFESSIONAL DEVELOPMENT

- Oct 2024 **Conference Presenter – ICCNet-2024, Manchester, United Kingdom**
“Dynamic Weight-Adjusted Ensemble Loss for Enhanced Medical Image Segmentation” – supported by the IoE International Travel Grant
- 2026 One-week Train-the-Trainer (T3) programme, *Cloud Application Development*, **IBM**
- 2025 One-week Train-the-Trainer (T3) programme, *Pattern Recognition*, **IBM**
- 2024 Workshop participant, *MRI and EEG Data Analysis*, IIT Hyderabad

TEACHING

COURSES TAUGHT

- **UPES Dehradun:** DevOps (B.Tech. Year 4, 2026); Research Methodology in Computer Science (B.Tech. Year 3, 2026); Python Programming (B.Tech. Year 1, 2026); Database Systems (M.Tech. Year 1, 2025); C Programming (B.Tech. Year 1, 2025)
- **Government Degree College Uri:** Data Structures and Algorithms; Discrete Mathematics (B.Sc. Computer Science, Year 2, 2020)

COURSES ASSISTED (TEACHING ASSISTANT, UNIVERSITY OF HYDERABAD)

- Deep Learning for Computer Vision
- Advanced Machine Learning
- Neural Networks and Applications
- Research Methodology

SUPERVISION AND MENTORING

- UPES: supervise B.Tech. and M.Tech. projects in machine learning and computer vision
- University of Hyderabad: mentored IMTech and M.Tech. students on course projects; guided two research assistants on fuzzy rough kernel-based extreme learning machines and mineral prospectivity classification
- Approach: project-based instruction that pairs mathematical foundations with reproducible implementation

TECHNICAL SKILLS

Programming	Python, C, MATLAB, SQL
Deep Learning	TensorFlow, Keras, PyTorch (working knowledge), CUDA / GPU training workflows
Machine Learning	Scikit-learn, NumPy, Pandas, SciPy, OpenCV, Matplotlib, Seaborn
Tools	Git, Docker, Jupyter, PyCharm, LaTeX, Linux / HPC, Mendeley, Zotero
Databases	MySQL
Domain	Medical image segmentation and classification, explainable AI, fuzzy and rough set methods, evolutionary optimisation

REFERENCES

Dr. Avatharam Ganivada

Doctoral advisor

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Further academic referees available on request.